POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2024
Programme: BE | Full Marks : 100
Course: Database Management System (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate between schema and instances. Explain data independence with its importance. 8
   b) Design an ER diagram for an Online Food Delivery System to manage customers, restaurants, orders, and delivery. Draw the ER diagram with cardinalities and constraints. 7
2. a) What is relational algebra? Explain Selection, Projection, Union, Cartesian-product, Join and Set-Difference operations with suitable examples. 7
   b) Consider the given relational schema: 8

   Movie(Title, Djrector, Myear, Rating)

   Actors(Actor, Age)

   Acts(Actor, Title)

   Directors (Director, DirAge)

   Write the SQL statement for the following task:

   i. Find the movies made by Gopal Varma after 1997.
   ii. Find all actors and directors.
   iii. Find number of movie which are directed by Devid Dhaban having actor "Bishnu".
   iv. Find (Director, Actor) pairs where the directors is younger than the actor.
   v. Find the number of movie actor wise.
3. a) Explain functional dependency with its types? For R ={A, B, C, D} and F={A->BC, B->C, AB->C, AC->D}, List the closure of functional dependency F. 7

OR

What are database constraints? Explain its types with suitable examples. When do we need Denormalization? Explain.

   b) Define normalization. Create a relation in Unnormalized Form (UNF) and normalize it up to the Second Normal Form (2NF). 8
4. a) Why security is needed in database? Explain access control mechanisms. 7

OR

As a database administrator, purpose the measures to protect the database against physical and cyber threats.

   b) Define ACID properties. Explain view serializability with suitable examples. 8
5. a) What do you mean by query processing? Explain the steps of query processing with necessary diagram. 7
   b) Construct a B+ tree for the following set of key values: {2,5,7,9,11,17,21,23,29,31}. Assume that the tree is initially empty and values are added in ascending order where the pointer number is FOUR. 8
6. a) Consider following two transactions T1 and T2 given in figure where T1 executes before T2. Also consider initial value of A, B and C are 1000, 2000 and 3000 respectively. 7

T1	T2
Read (A)	Read (C)
A = A - 500	C = C – 200
Write (A)	Write (C)
Read (B)	
B = B + 500	
Write (B)	

Use the concept of deferred and immediate database modification log based recovery method in given problem.

   b) Define data warehousing and explain how it supports decision-making processes in large organizations. 8
7. Write short notes on: **(Any two)** 2×5
   a) Query cost estimation
   b) Set operations in SQL
   c) NoSQL Databases and its characteristics

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